

Lecture 3

Introduction to OpAmp

Review

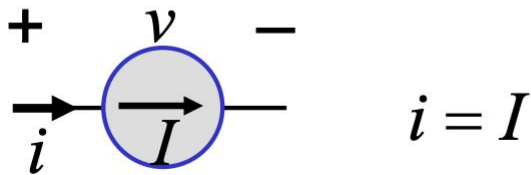
- Last class: alternative representation, KCL, KVL, nodal
- Today: **review** of dependent source, introduction to amplifiers, operational amplifiers

Dependent Source

- Active sources – either voltage or current source
- Difference: voltage (or current) controlled by the voltage/current in a different branch
- Technically non-linear devices – because in real life, dependent sources are made using non-linear electronic devices (e.g. MOSFET)
- However, analysis are similar to linear circuits (CSE250), hence we will start with this

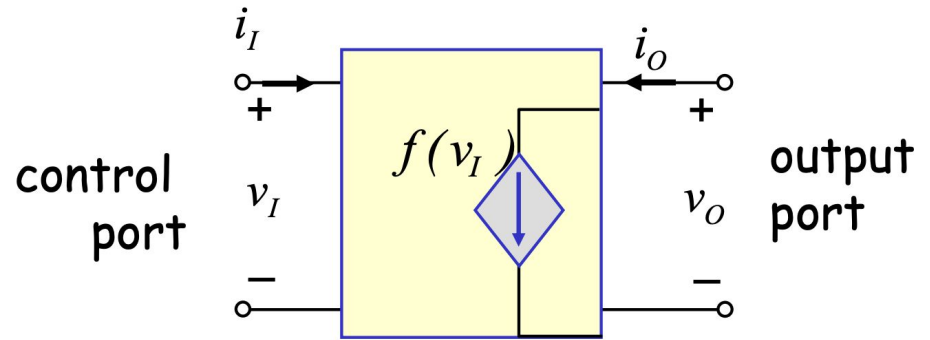
Major Difference

Independent devices:



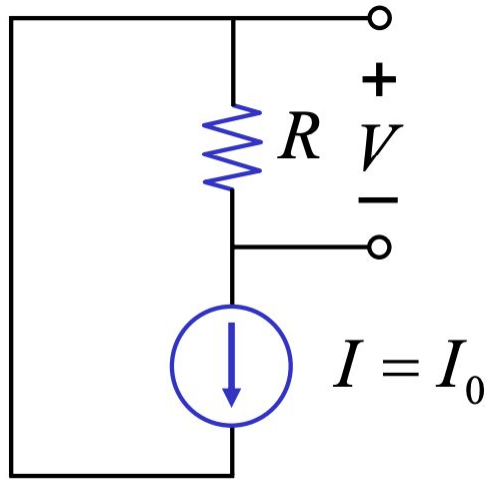
2 terminals, 1 port

Dependent devices:



2 terminals, 2 port

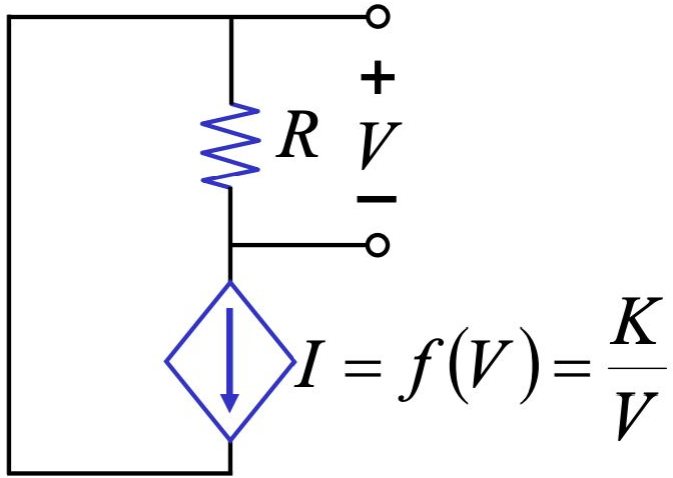
Example 1



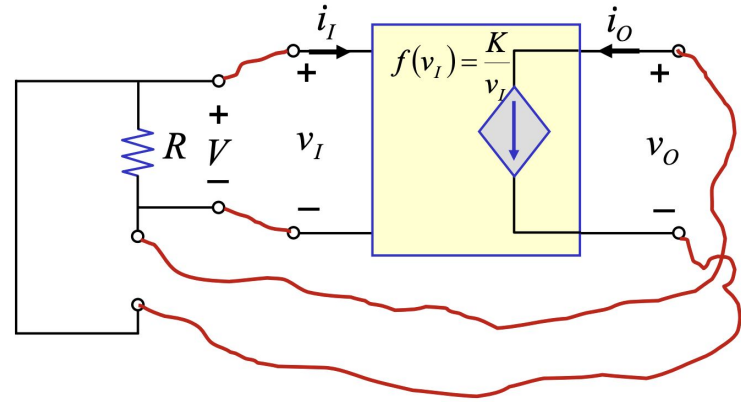
Find V

(Ans: $V = I_0 R$)

Example 2



Find V

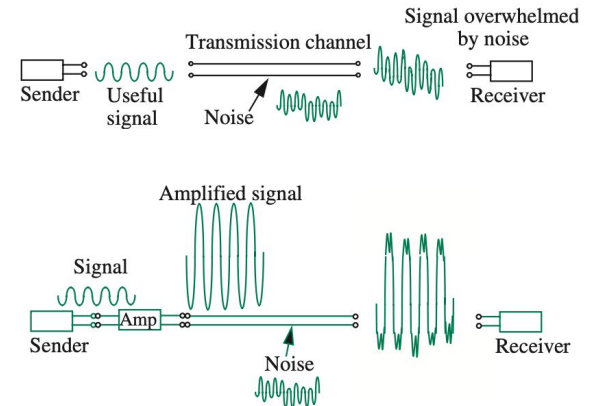
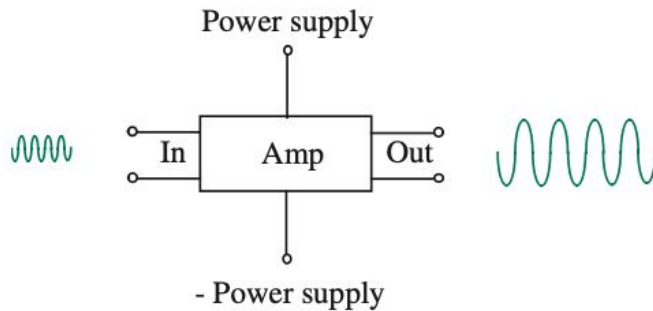


$$\begin{aligned} \text{Ans: } V &= IR \Rightarrow V = \left(\frac{K}{V}\right) R \\ &\Rightarrow V^2 = KR \Rightarrow V = \sqrt{KR} \end{aligned}$$

Application of dependent sources: amplifiers!

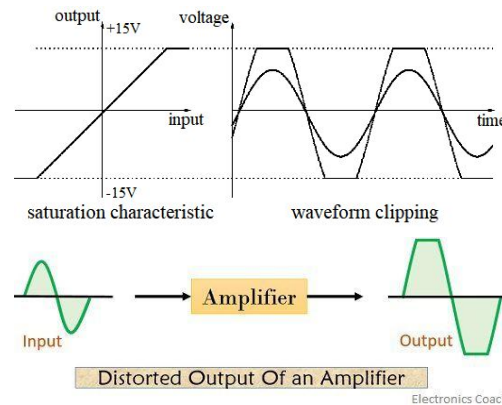
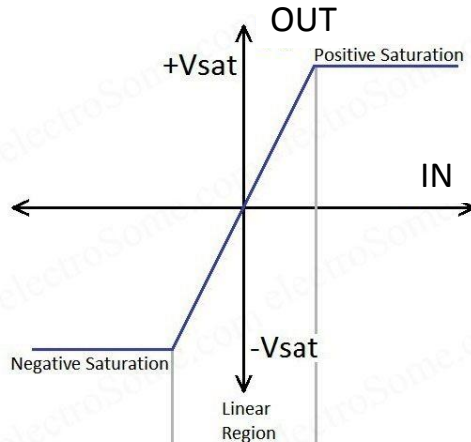
Amplifier

- Linear amplifier $\Rightarrow OUT = k \times IN$. k is called the **gain** of the amplifier
- OUT/IN can be voltage or current
- Amplifier must provide power gain ($P_{OUT} > P_{IN}$), hence need **power supply**
- Usage: communication



Transfer Characteristics of Amplifiers

- $y = mx$, hence straight line going through origin. Here, $m = k = \text{gain}$
- Saturation due to limited power supply. Output cannot be greater than P.S.
- Input must be within a **valid input range**, otherwise output will be distorted
- Amplifiers are **non-linear devices**, because they are made using dependent sources

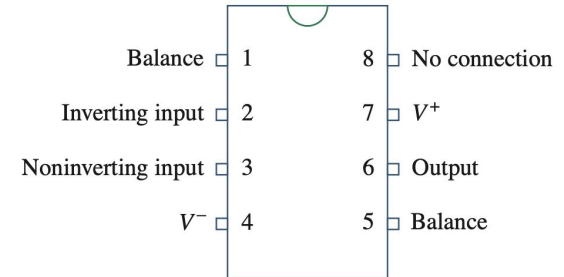
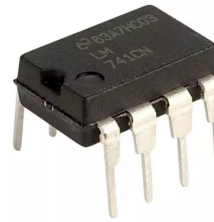
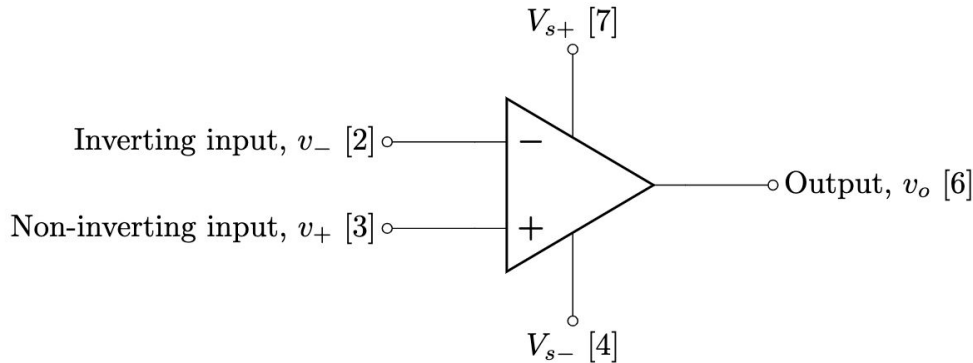


Operational Amplifiers

- An operational amplifier, or **op-amp** for short, is a versatile and powerful integrated circuit that is widely used in a variety of electronic applications.
- An Op-Amp is designed so that it performs some mathematical operations when external components, such as resistors and capacitors, are connected to its terminals.
- The op amp is an electronic device consisting of a complex arrangement of resistors, transistors, capacitors, and diodes. A full discussion of what is inside the op amp is beyond the scope of this course. For now, it will suffice to treat the op amp as a circuit building block and simply study what takes place at its terminals.

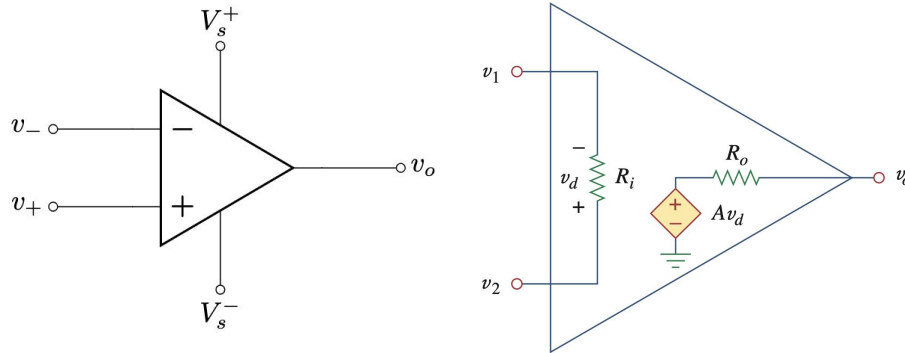
Terminals and Circuit Symbol

- Op-amps are **differential amplifiers** – meaning it amplifies the voltage difference between two terminals. Therefore, it has two inputs
- Since this is an amplifier, there are two terminals for **power supply**



Equivalent Circuit

Since op amp is an amplifier, the internal circuit can be modeled using a **voltage controlled voltage source (VCVS)**! (actual circuit is complicated)



$v_1 = v_-$ = voltage of inverting terminal

$v_2 = v_+$ = voltage of noninverting terminal

$v_d = v_+ - v_- = v_2 - v_1$
= differential input voltage for VCVS

A = Open loop gain

R_i = Input resistance

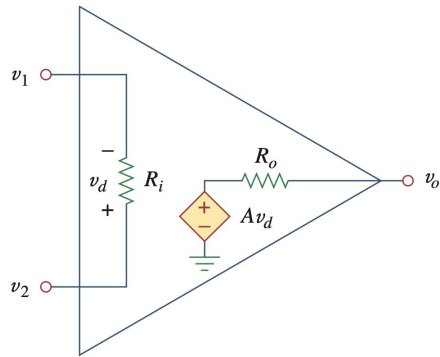
R_o = Output resistance

The op amp senses the difference between the two inputs, multiplies it by the gain A , and causes the resulting voltage to appear at the output. Thus, the output v_o is given by

$$v_o = Av_d = A(v_2 - v_1) = A(v_+ - v_-)$$

Equivalent Circuit

Since op amp is an amplifier, the internal circuit can be modeled using a **voltage controlled voltage source (VCVS)**! (actual circuit is complicated)



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Parameter	Typical Range

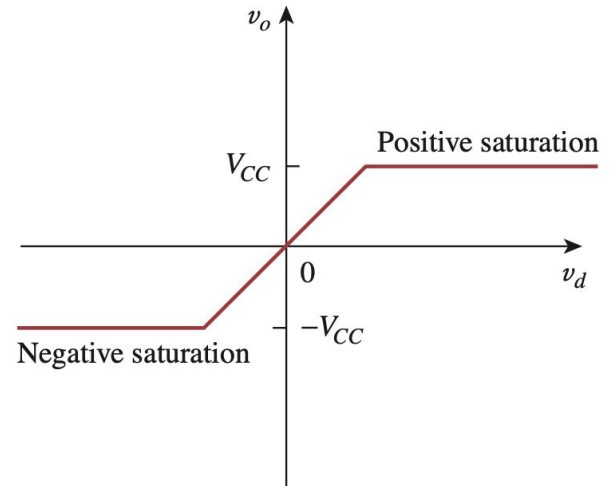
The op amp senses the difference between the two inputs, multiplies it by the gain A , and causes the resulting voltage to appear at the output. Thus, the output v_o is given by

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Practical Limitation - Saturation

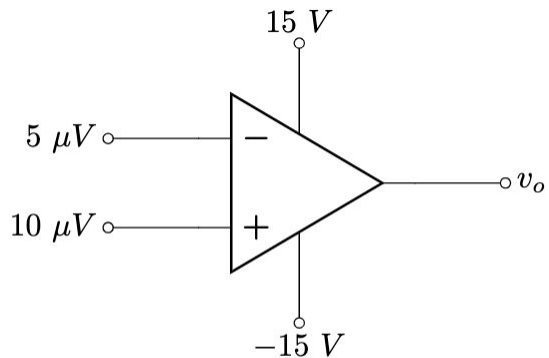
- Similar to amplifiers, the output of op-amp is limited by power supply (Saturation)
- Op-amp can operate in three modes, depending on v_d :
 - Positive saturation: $v_o = V_s^+ = V_{CC}$
 - Linear region: $-V_{CC} < v_o = Av_d < V_{CC}$ [Slope = A]
 - Negative saturation: $v_o = V_s^- = -V_{CC}$
- If we attempt to increase v_d beyond the linear range, the op-amp becomes saturated and yields $v_o = V_{CC}$ or $v_o = -V_{CC}$. That is the output voltage is restricted by

$$-V_{CC} \leq v_o \leq V_{CC}$$



Example 3

- Find v_o



$$A = 2 \times 10^5$$

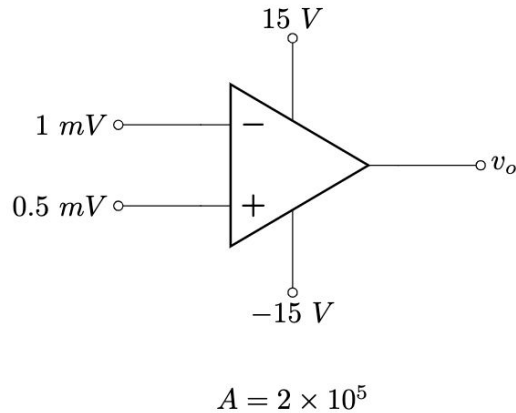
Solution:

$$\begin{aligned} v_d &= v_+ - v_- = 10\ \mu V - 5\ \mu V = 5\ \mu V \\ \Rightarrow v_o &= Av_d = (2 \times 10^5) \times (5 \times 10^{-6}) = 1\ V \end{aligned}$$

Since this is in between $-V_{CC}$ and V_{CC} , output valid

Example 4

- Find v_o



Solution:

$$v_d = v^+ - v^- = 0.5 \text{ mV} - 1 \text{ mV} = -0.5 \text{ mV}$$
$$\rightarrow v_o = Av_d = (2 \times 10^5) \times (-0.5 \times 10^{-3}) = -100 \text{ V}$$

However, the output must be limited within the range of $-V_{CC}$ to $+V_{CC}$. Therefore, the highest output voltage can be -15 V .

$$\therefore v_o = -15 \text{ V}$$

Solving Circuit with Op-Amp

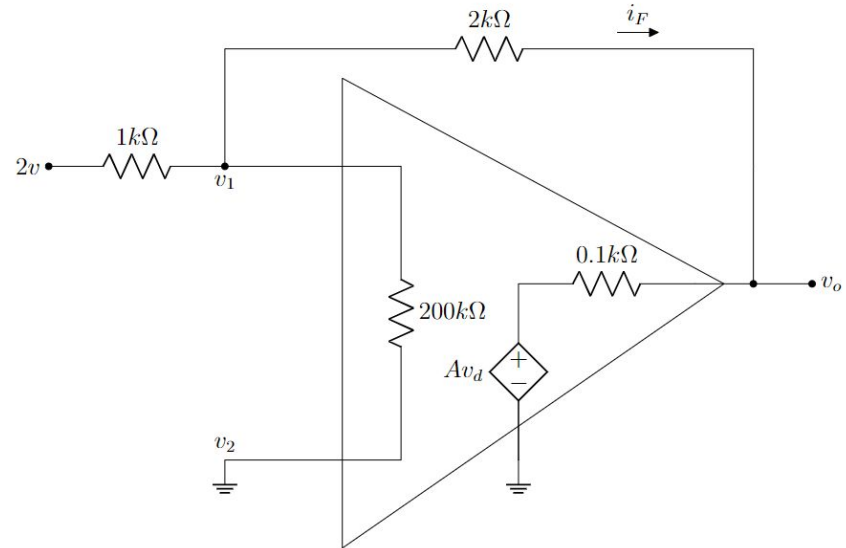
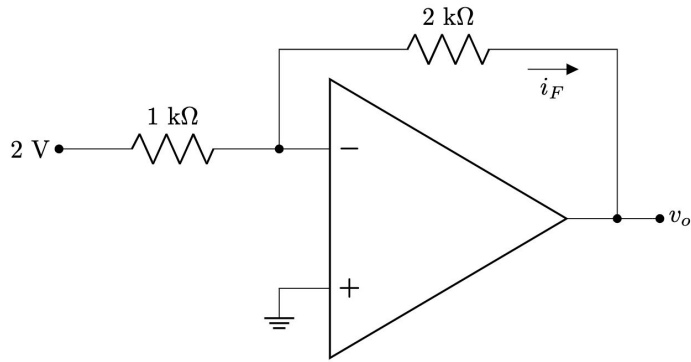
- Don't be scared! Circuit solving with op-amp is exactly same as the circuits you solved in CSE250
- Step 1: replace with equivalent circuit
- Step 2: solve using KCL & KVL, or nodal analysis (linear)
- In most cases, the power supply voltages are not mentioned. **Hence, we will assume the output is within the linear range in this case**

Example 5

Find i_F and v_o . Here, $R_i = 200\text{ k}\Omega$, $R_o = 0.1\text{ k}\Omega$, $A = 2 \times 10^5$

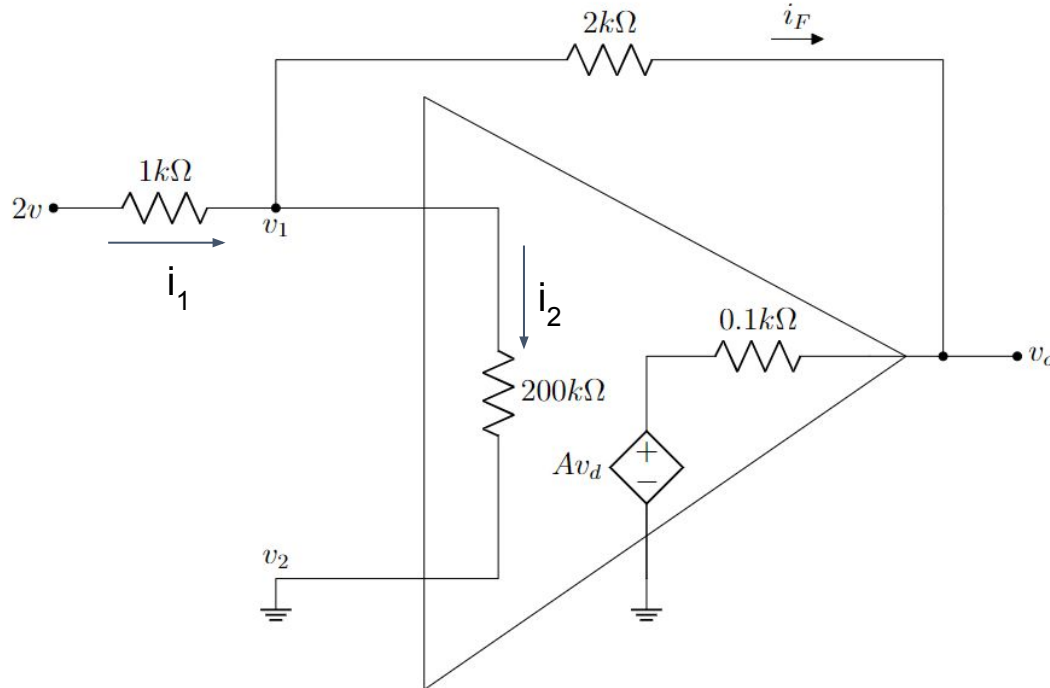
Solution:

Step 1: replace using equivalent circuit



Example 5

Step 2: Solve using KCL & KVL or nodal



Applying KCL we get,

$$i_1 = i_2 + i_F$$

$$\rightarrow (2-v_1)/(1k) = (v_1-v_2)/(200k) + (v_1-v_o)/(2k)$$

$$\rightarrow (2-v_1)/(1k) = (v_1-0)/(200k) + (v_1-v_o)/(2k)$$

$$\rightarrow (2-v_1)/(1k) = v_1/(200k) + (v_1-v_o)/(2k) \dots \dots (i)$$

Again, from the figure,

$$i_{0.1k} = i_F$$

$$\rightarrow (v_o - Av_d)/(0.1k) = (v_1-v_o)/(2k)$$

$$\rightarrow (v_o - (2 \times 10^5) \times (v_2 - v_1))/(0.1k) = (v_1-v_o)/(2k)$$

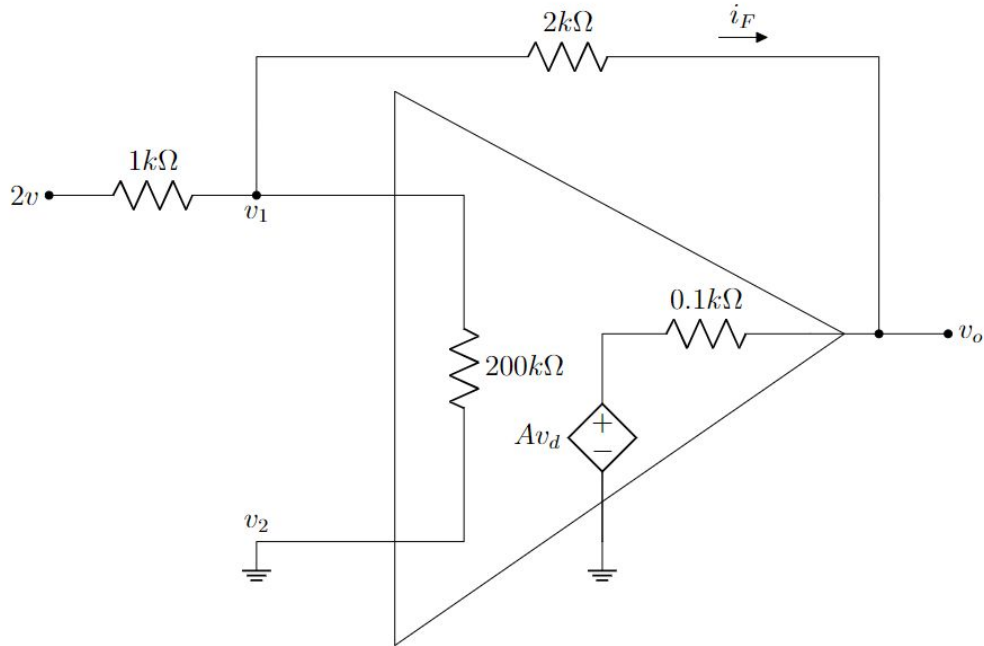
$$\rightarrow (v_o - (2 \times 10^5) \times (0 - v_1))/(0.1k) = (v_1-v_o)/(2k)$$

$$\rightarrow (v_o - (2 \times 10^5) \times (-v_1))/(0.1k) = (v_1-v_o)/(2k) \dots (ii)$$

We can get v_1 and v_o by solving equation (i) and (ii)

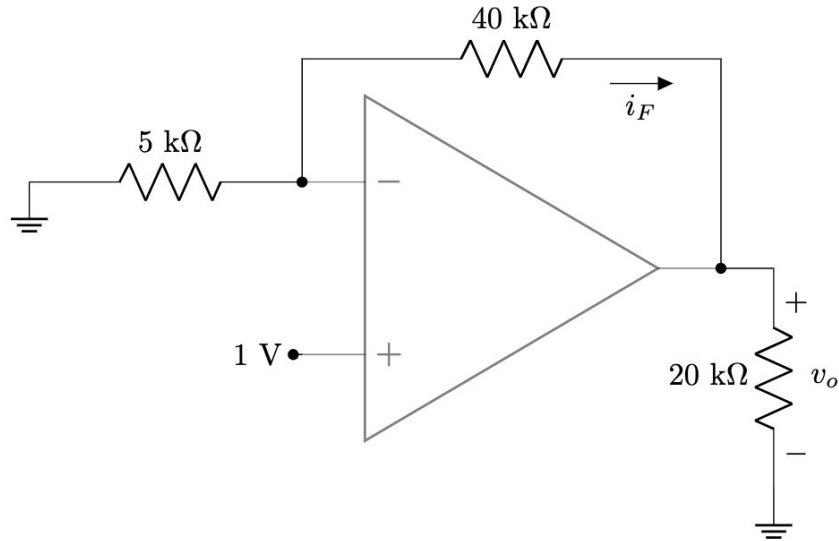
Example 5

Step 2: Solve using KCL & KVL or nodal



Example 6

Find i_F and v_o . Here, $R_i = 200\text{ k}\Omega$, $R_o = 0.1\text{ k}\Omega$, $A = 2 \times 10^5$

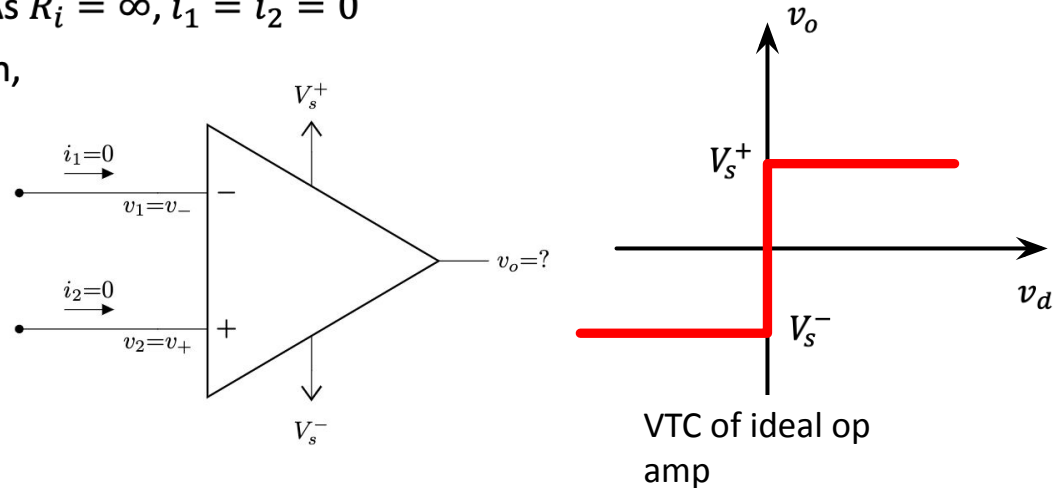


The Ideal Op-Amp

- To facilitate the understanding of op amp circuits, we will assume **ideal op amps**
 - Infinite open-loop gain, $A = \infty$
 - Infinite input resistance, $R_i = \infty =$ open circuit
 - Zero output resistance, $R_o = 0 =$ short circuit
- Although an ideal op amp provides only an approximate analysis, most modern amplifiers have such large gains and input impedances that the approximate analysis is a good one.
- **Circuit solving become much simpler.** As $R_i = \infty, i_1 = i_2 = 0$
- Since $A = \infty$, in open-loop configuration, v_o will either be positive saturated or negative saturated (why?)

$$v_o = \begin{cases} V_s^+ & \text{if } v_d > 0 \Rightarrow v_2 > v_1 \\ V_s^- & \text{if } v_d < 0 \Rightarrow v_2 < v_1 \end{cases}$$

Reminder: $v_d = v_2 - v_1 = v_+ - v_-$



Application - Comparator

- A comparator compares two voltages to determine which is larger.
- The comparator is essentially an op-amp operated in an open-loop configuration
- Two types –
 - (1) **Non-inverting**: outputs a positive voltage ($V_H = V_S^+$) when input is greater than reference
 - (2) **Inverting**: outputs a negative voltage ($V_L = V_S^-$) when input is greater than reference
- Application – smoke detector, turning AC on/off automatically, etc (next lecture)

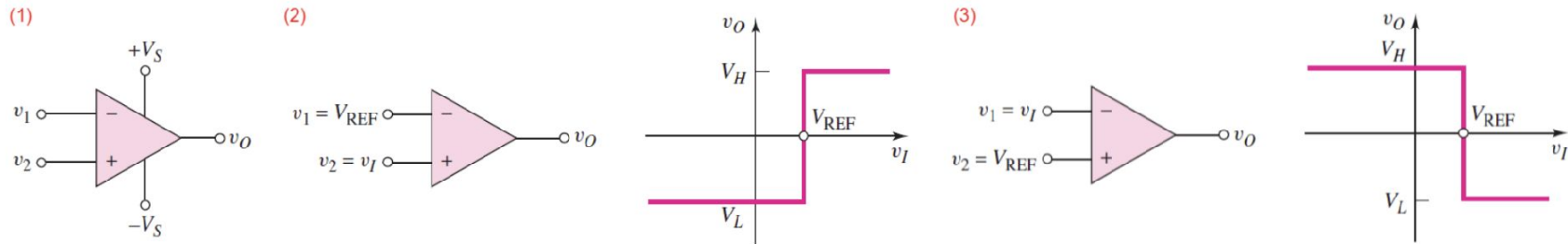
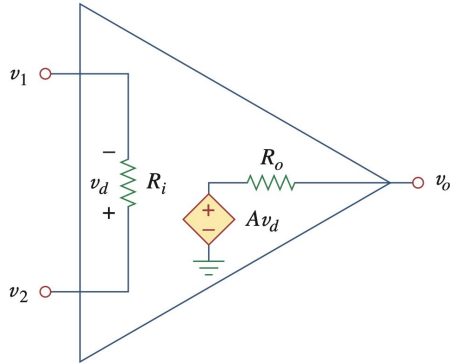
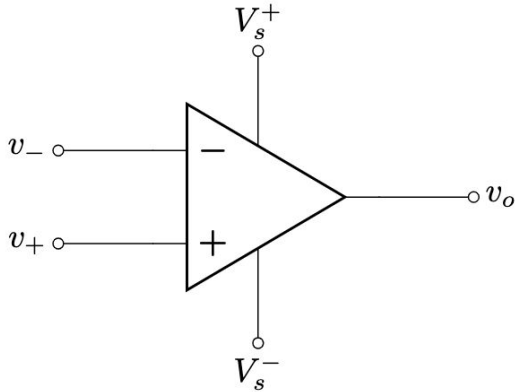


Figure 2: (1) Op-Amp Comparator (2) Noninverting Circuit (3) Inverting Circuit

Lecture 4

Op Amp – Part 2

Review – Operational Amplifier



$v_1 = v_-$ = voltage of inverting terminal

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$v_d = v_+ - v_- = v_2 - v_1$
= differential input voltage for VCVS

A = Open loop gain

R_i = Input resistance

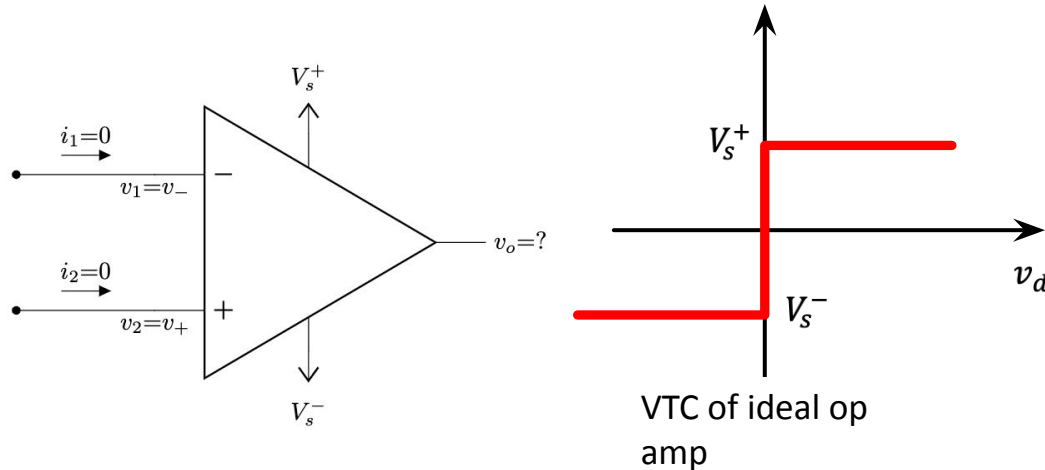
R_o = Output resistance

Differential amplifier $\Rightarrow$ amplifies the difference

$$v_o = A v_d = A(v_2 - v_1) = A(v_+ - v_-)$$

Review – Ideal Op Amp

- Infinite open-loop gain, $A = \infty$
- Infinite input resistance, $R_i = \infty = \text{open circuit}$
- Zero output resistance, $R_o = 0 = \text{short circuit}$
- As $R_i = \infty$ (open circuit), $i_1 = i_2 = 0$. Therefore, circuit solving become much simpler



$$v_o = \begin{cases} V_s^+ & \text{if } v_d > 0 \Rightarrow v_2 > v_1 \\ V_s^- & \text{if } v_d < 0 \Rightarrow v_2 < v_1 \end{cases}$$

Application of Ideal Op Amp - Comparator

- A comparator compares two voltages to determine which is larger.
- The comparator is essentially an op-amp operated in an open-loop configuration
- Two types –
 - (1) **Non-inverting**: outputs a positive voltage ($V_H = V_S^+$) when input is greater than reference
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- Application – smoke detector, turning AC on/off automatically, etc

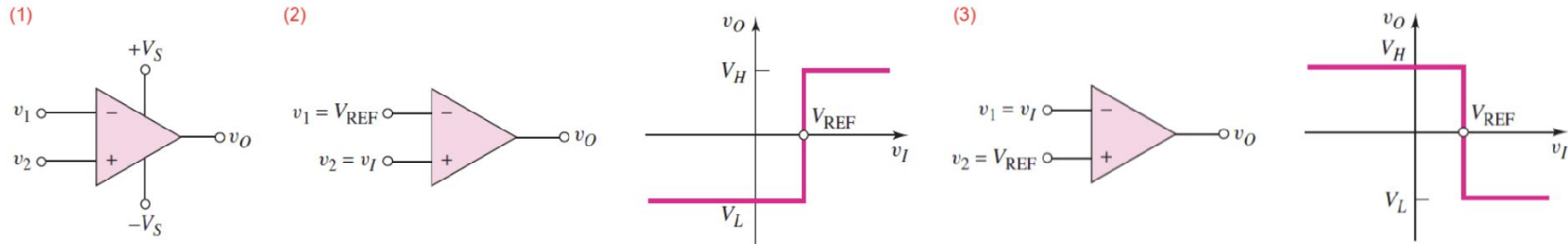


Figure 2: (1) Op-Amp Comparator (2) Noninverting Circuit (3) Inverting Circuit

Comparator Application – Automatic AC

Auto AC ON/OFF

Sensor



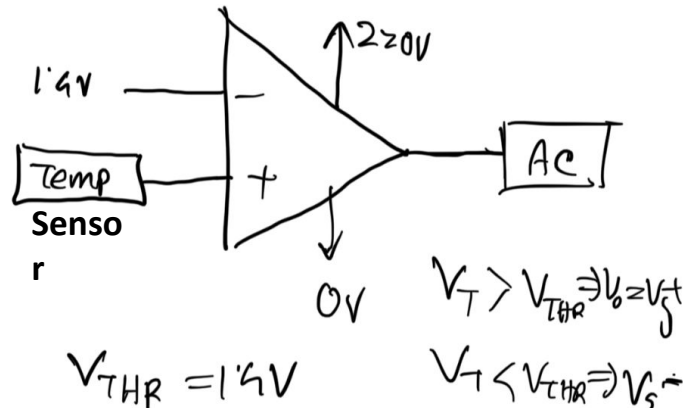
$$V_T \propto T$$

$$23^\circ, V_T = 1.2V$$

$$24^\circ, V_T = 1.4V$$

$$25^\circ, V_T = 1.6V$$

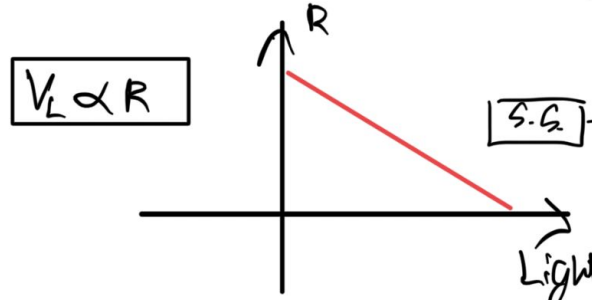
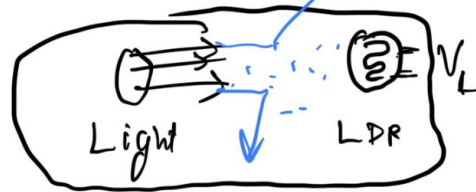
AC should be on if
 $T > 24^\circ C$



Smoke Detector

③ Smoke Detector

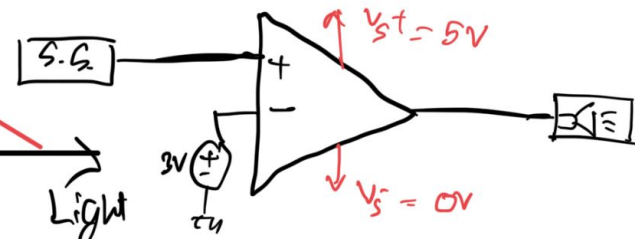
Smoke sensor



No smoke, $V_L = \text{small} = 2V$

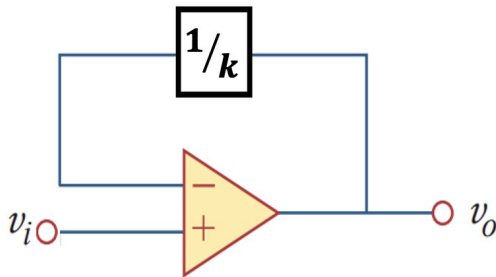
Smoke, $V_L = \text{high} = 4V$

$V_L > 3V \Rightarrow \text{Alarm ON}$
 $V_L < 3V \Rightarrow \text{OFF}$



Introducing Negative Feedback

- The gain (A) of an ideal op amp is infinity, practically extremely large.
- The power supply ($+V_s$ and $-V_s$) limits the op amp's output.
- We require a method to have a finite gain. That is what negative feedback does.
- Negative feedback: feeding back **a portion** of output to inverting input
- Idea – the output will become stable due to a self-correcting mechanism

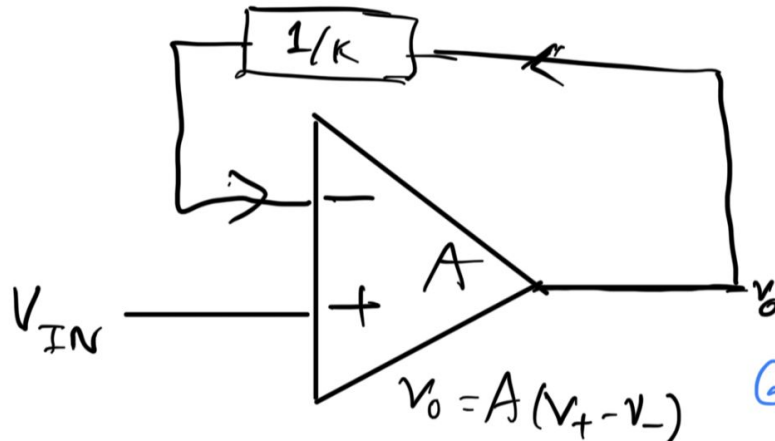


For example, here, $v_- = \text{one } k'\text{th part of output} = \frac{1}{k}$

If v_o increases, v_- will increase, hence v_d will decrease, eventually v_o decreases

If v_o decreases, v_- will decrease, hence v_d will increase, eventually v_o increase

Negative Feedback – Numerical Example

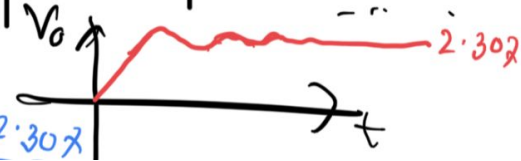


$$\underline{V_{IN} = 1V}, \quad K = 10, \quad A = 3$$

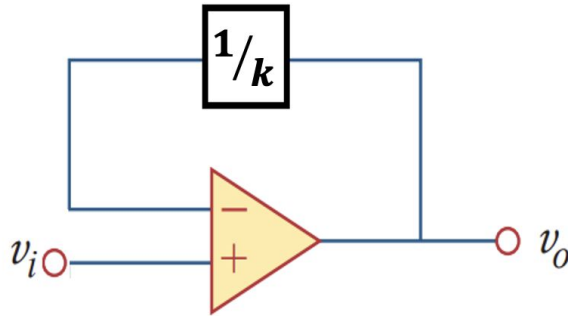
$$\begin{aligned}
 V_O &= 0V \\
 V_- &= 0V \\
 V_d &= 1V \\
 \therefore V_O &= 3V
 \end{aligned}$$

$V_O = 3V$	$V_O = 2.1V$	$V_O = 2.302$
$V_- = 0.3V$	$V_- = 0.21V$	$V_- = 0.2302$
$V_d = 0.7$	$V_d = 0.29$	$V_d = 0.2698$
$V_O = 2.1V$	$V_O = 2.32$	$V_O = 2.302$

$$Gain = \frac{V_O}{V_{IN}} = 2.302$$



Negative Feedback – Derivation of Gain



$$\text{Here, } v_- = \frac{v_o}{k}$$

$$\text{We know, } v_o = A v_d$$

$$\text{or, } v_o = A(v_+ - v_-)$$

$$= A\left(v_i - \frac{v_o}{k}\right)$$

$$= A v_i - \frac{A}{k} v_o$$

$$\text{or, } v_o\left(1 + \frac{A}{k}\right) = A v_i$$

$$\text{So, } v_o = \frac{A v_i}{1 + \frac{A}{k}}$$

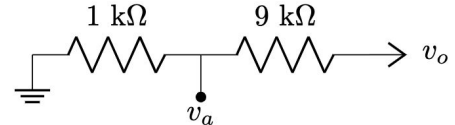
$$\text{or, } v_o = \frac{v_i}{\frac{1}{A} + \frac{1}{k}}$$

A is extremely large,
so, $\frac{1}{A} \approx 0$

$$v_o = \frac{v_i}{\frac{1}{k}} = k v_i$$

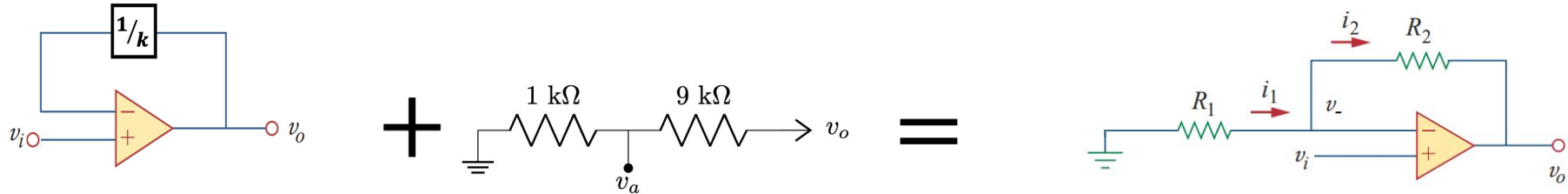
If $k = 10$ (meaning we feed back one tenth of the output to negative input), we will get $v_o = 10 v_i$. that is 10 fold gain.

How to get $1/k$ of output to input?
Voltage dividers!



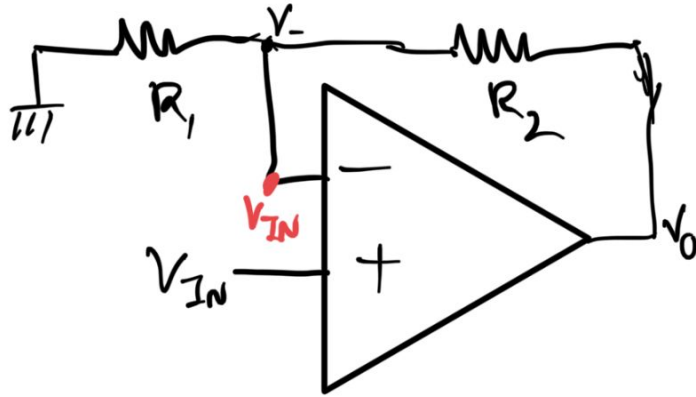
$$v_a = \frac{1 \text{ k}\Omega}{1 \text{ k}\Omega + 9 \text{ k}\Omega} \times v_o = \frac{v_o}{10}$$

Inverting Amplifier



Inverting Amplifier

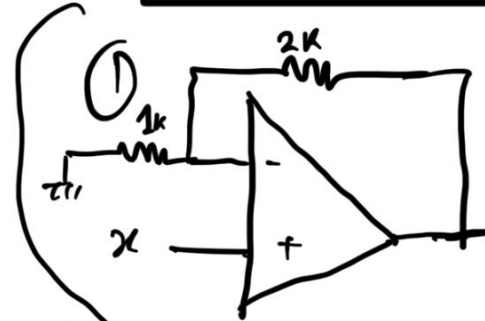
Gain > 1



$$\therefore V_O = \left(1 + \frac{R_2}{R_1}\right) V_{IN}$$

$$V_- = \frac{R_1}{R_1 + R_2} V_O \approx \frac{1}{1 + \frac{R_2}{R_1}} V_O$$

K

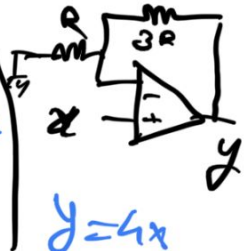


$$y = \left(1 + \frac{2}{1}\right) x$$

$$\Rightarrow y = 3x$$

② Design $y = 4x$

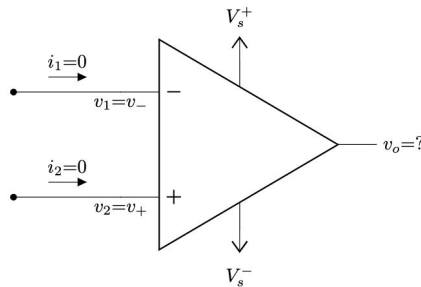
$$1 + \frac{R_2}{R_1} = 4 \Rightarrow \frac{R_2}{R_1} = 3$$



$$y = 4x$$

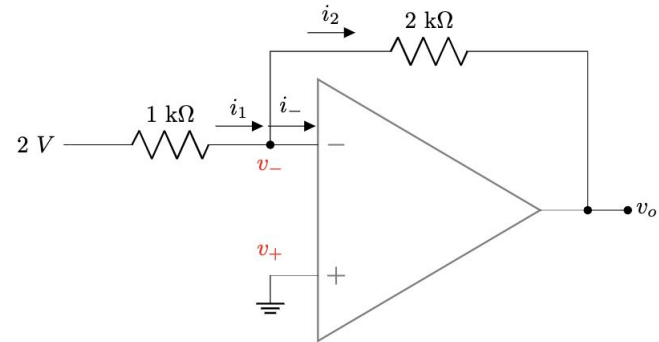
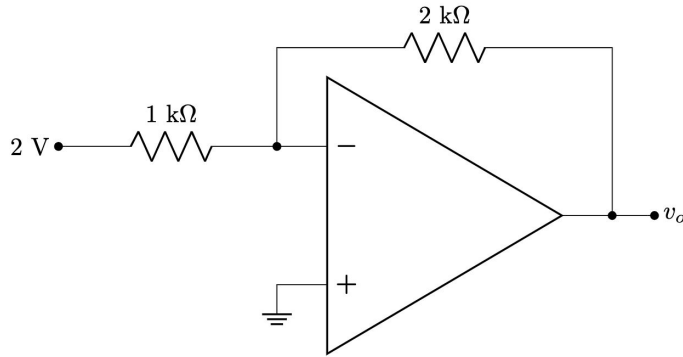
Solving Circuit with Ideal Op Amp + NF

- For ideal op-amp
 - Infinite input resistance, $R_i = \infty = \text{open circuit}$
 - Zero output resistance, $R_o = 0 = \text{short circuit}$
 - $i_i = 0$ and $i_+ = 0$
- **When there is negative feedback**, For ideal A as is infinitely high, for a finite output voltage v_o , $\frac{v_o}{A} = v_d = 0 \Rightarrow v_+ = v_-$. This is called **virtual short circuit**
- Because of these, solving ideal op-amp circuit with negative feedback is very simple

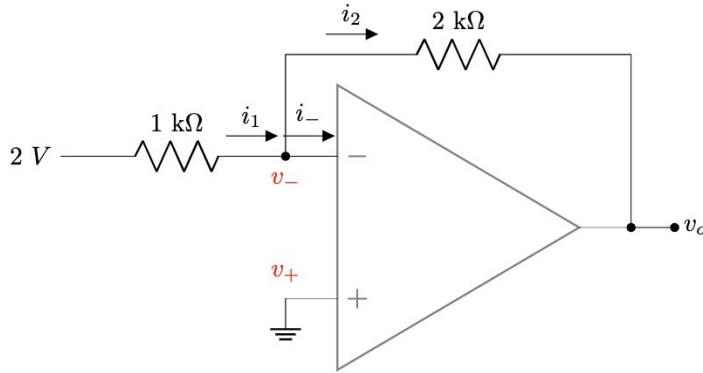


Example – Inverting Amplifier

Solve the circuit to find v_o



Example – Inverting Amplifier



Since v_+ is connected to ground, $v_+ = 0V$

Since there is negative feedback, from virtual short, $v_- = v_+ = 0V$

From Ohm's law for $1\text{ k}\Omega \Rightarrow i_1 = \frac{2V - 0V}{1\text{ k}\Omega} = 2\text{mA}$

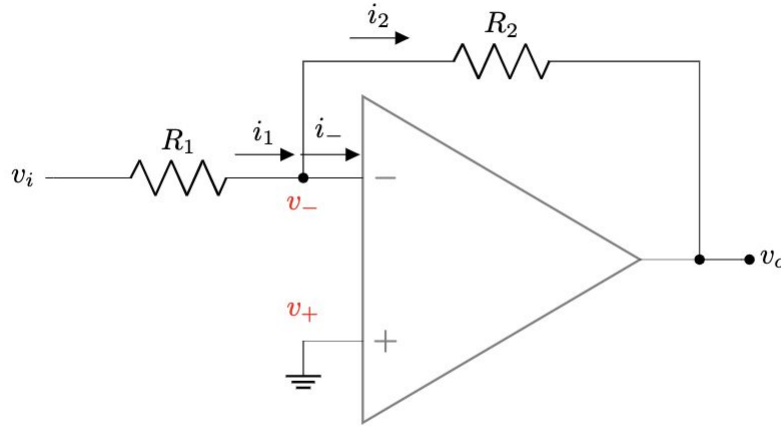
$$\text{Gain} = -\frac{4V}{2V} = -2 \text{ (hence **inverting**)}$$

Since ideal op-amp, $i_- = i_+ = 0$

From KCL at v_- , $i_1 = i_- + i_2 \Rightarrow i_1 = i_2 = 2\text{mA}$

From Ohm's law for $2\text{ k}\Omega \Rightarrow i_2 = \frac{v_- - v_o}{2\text{ k}\Omega} = 2\text{mA} \Rightarrow v_o = -i_2 \times 2 = -4V \text{ [ANS]}$

General



Since v_+ is connected to ground, $v_+ = 0V$

Since there is negative feedback, from virtual short, $v_- = v_+ = 0V$

From Ohm's law for $R_1 \Rightarrow i_1 = \frac{v_i - 0V}{R_1} = v_i/R_1$

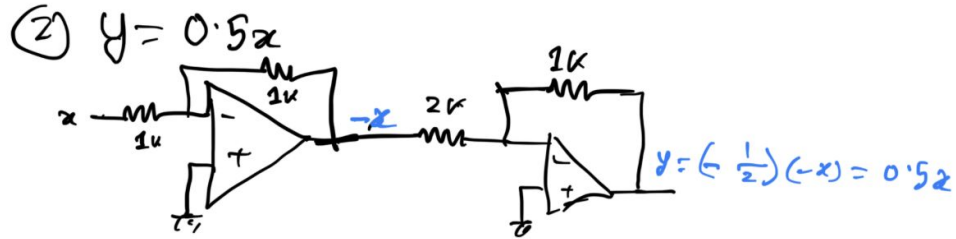
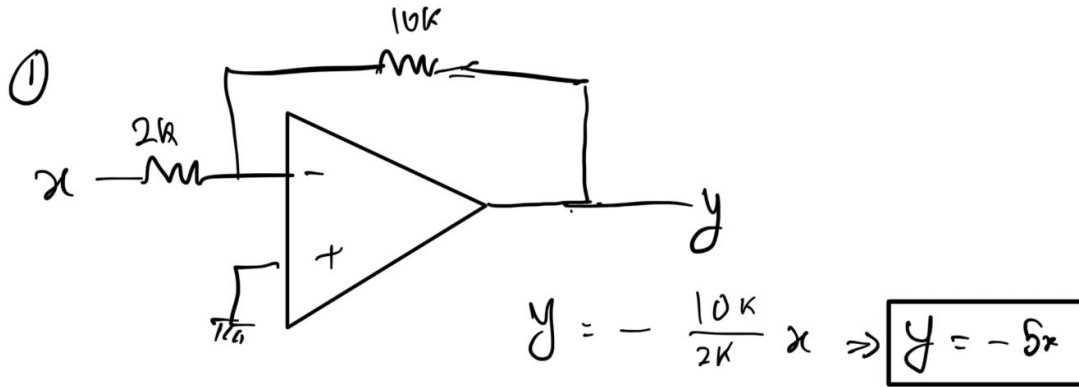
Since ideal op-amp, $i_- = i_+ = 0$

From KCL at v_- , $i_1 = i_- + i_2 \Rightarrow i_1 = i_2 = v_i/R_1$

From Ohm's law for $R_2 \Rightarrow i_2 = \frac{v_- - v_o}{R_2} = \frac{v_i}{R_1} \Rightarrow v_o = -i_2 \times R_2 \Rightarrow v_o = -\frac{R_2}{R_1} v_i$ [ANS]

$$\text{Gain} = -\frac{R_2}{R_1}$$

Example



Lecture 5

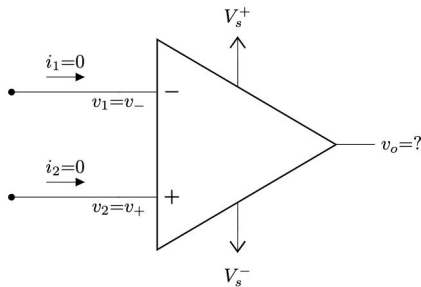
Op Amp – Part 3

Solving Circuit with Ideal Op Amp + NF

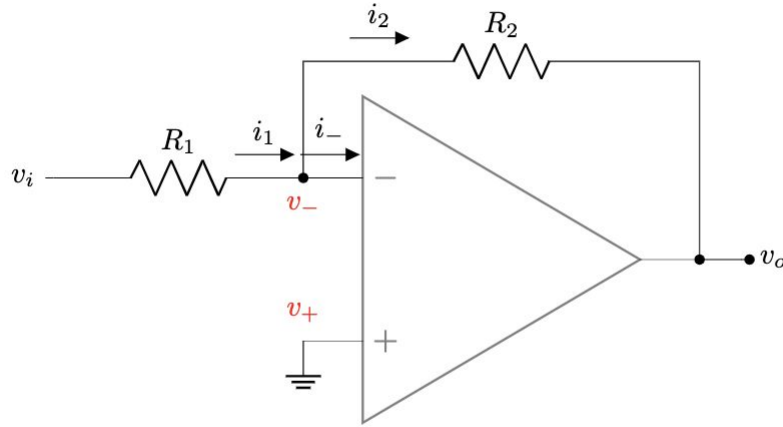
- For ideal op-amp

- Infinite input resistance, $R_i = \infty = \text{open circuit}$
- Zero output resistance, $R_o = 0 = \text{short circuit}$
- $i_i = 0$ and $i_+ = 0$

- **When there is negative feedback**, For ideal A as is infinitely high, for a finite output voltage v_o , $\frac{v_o}{A} = v_d = 0 \Rightarrow v_+ = v_-$. This is called **virtual short circuit**
- Because of these, solving ideal op-amp circuit with negative feedback is very simple



Review – Inverting Amplifier



Since v_+ is connected to ground, $v_+ = 0V$

Since there is negative feedback, from virtual short, $v_- = v_+ = 0V$

From Ohm's law for $R_1 \Rightarrow i_1 = \frac{v_i - 0V}{R_1} = v_i/R_1$

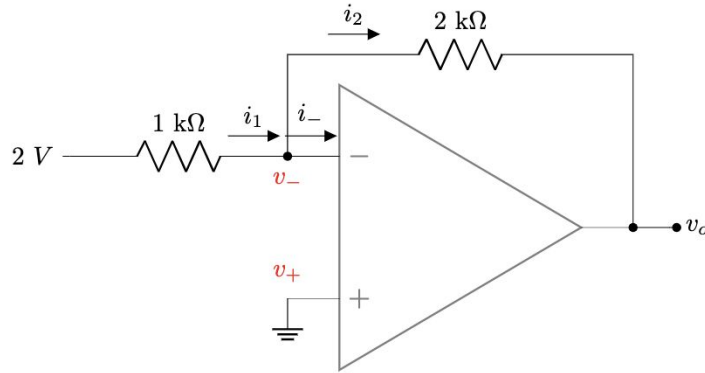
Since ideal op-amp, $i_- = i_+ = 0$

From KCL at v_- , $i_1 = i_- + i_2 \Rightarrow i_1 = i_2 = v_i/R_1$

From Ohm's law for $R_2 \Rightarrow i_2 = \frac{v_- - v_o}{R_2} = \frac{v_i}{R_1} \Rightarrow v_o = -i_2 \times R_2 \Rightarrow v_o = -\frac{R_2}{R_1} v_i$ [ANS]

$$\text{Gain} = -\frac{R_2}{R_1}$$

Example – Inverting Amplifier



Since v_+ is connected to ground, $v_+ = 0V$

Since there is negative feedback, from virtual short, $v_- = v_+ = 0V$

From Ohm's law for 1 kΩ $\Rightarrow i_1 = \frac{2V - 0V}{1 \text{ k}\Omega} = 2mA$

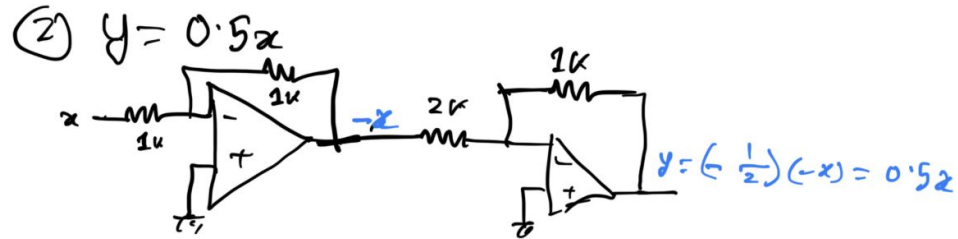
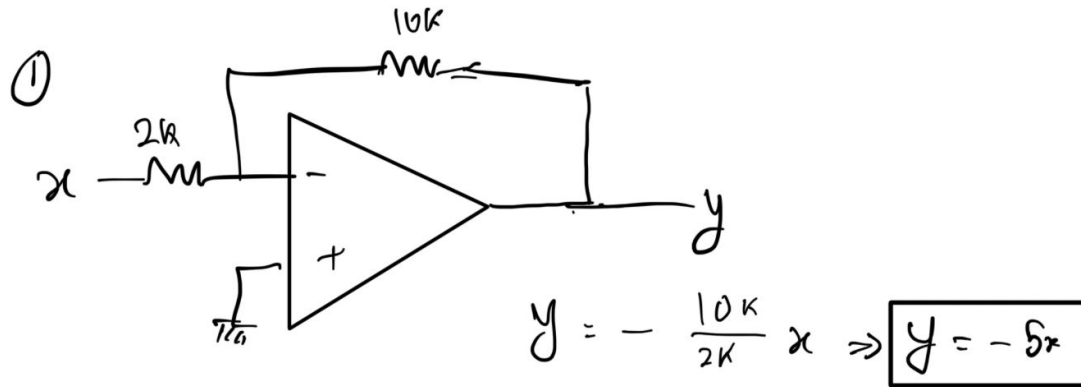
$$\text{Gain} = -\frac{4V}{2V} = -2 \text{ (hence **inverting**)}$$

Since ideal op-amp, $i_- = i_+ = 0$

From KCL at v_- , $i_1 = i_- + i_2 \Rightarrow i_1 = i_2 = 2 \text{ mA}$

From Ohm's law for 2 kΩ $\Rightarrow i_2 = \frac{v_- - v_o}{2 \text{ k}\Omega} = 2mA \Rightarrow v_o = -i_2 \times 2 = -4V$ [ANS]

Example



Inverting Adder

Consider v_1 first, and deactivate other (v_2, v_3, v_4) sources.

It is nothing but a non-inverting amplifier.

$$\text{So, } v_{o1} = -\frac{R_f}{R_1} v_1$$

Similarly, if we active one source and deactivate others, we will get:

$$v_{o2} = -\frac{R_f}{R_2} v_2, v_{o3} = -\frac{R_f}{R_3} v_3, v_{o4} = -\frac{R_f}{R_4} v_4$$

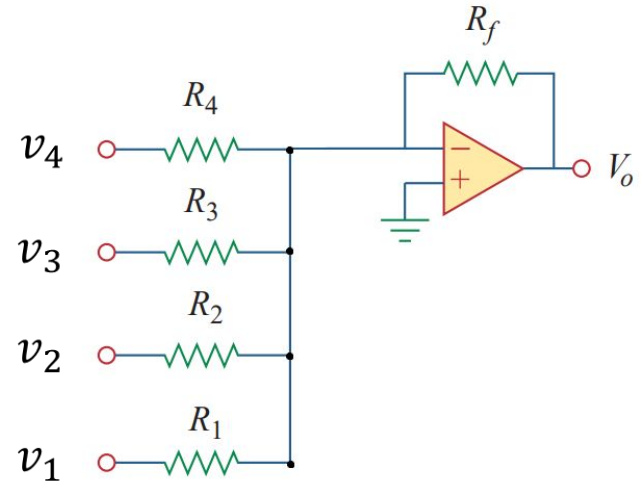
Now, using **superposition principle**,

$$v_o = v_{o1} + v_{o2} + v_{o3} + v_{o4}$$

$$\text{So, } v_o = -\frac{R_f}{R_1} v_1 - \frac{R_f}{R_2} v_2 - \frac{R_f}{R_3} v_3 - \frac{R_f}{R_4} v_4$$

$$\text{Or, } v_o = -\left(\frac{R_f}{R_1} v_1 + \frac{R_f}{R_2} v_2 + \frac{R_f}{R_3} v_3 + \frac{R_f}{R_4} v_4\right)$$

We can use this circuit to add any 'n' number of inputs!



Example

Implement the following function using op-amps:

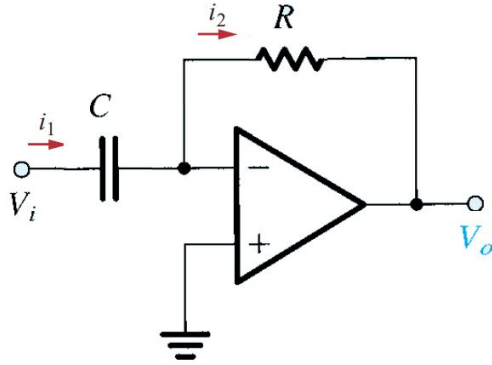
$$v_0 = -(v_1 + 0.5v_2 + v_3)$$

Solution:

Here, $R_f/R_1 = 1$, $R_f/R_2 = 0.5$, $R_f/R_3 = 1$

If $R_f = 1 \text{ k}\Omega$, $R_2 = 2 \text{ k}\Omega$, $R_3 = 1 \text{ k}\Omega$

Op Amp as Differentiator



Since v_+ is connected to ground, $v_+ = 0V$

Since there is negative feedback, from virtual short, $v_- = v_+ = 0V$

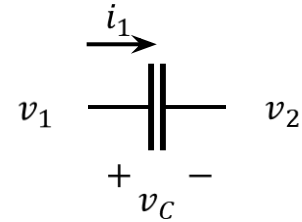
For the capacitor C , $\Rightarrow i_1 = C \frac{dv_C}{dt} = C \frac{d(v_i - v_-)}{dt} = C \frac{dv_i}{dt}$

From Ohm's law for $R \Rightarrow i_2 = \frac{v_- - v_o}{R} = -\frac{v_o}{R}$

Since ideal op-amp, $i_- = i_+ = 0$, so $i_1 = i_2$

$$\Rightarrow -\frac{v_o}{R} = C \frac{dv_i}{dt} \Rightarrow v_o = -RC \frac{dv_i}{dt} \text{ [Ans.]}$$

Review – Capacitor



$$i_1 = C \frac{dv_C}{dt} = C \frac{d(v_1 - v_2)}{dt}$$

Op Amp as Integrator

Since v_+ is connected to ground, $v_+ = 0V$

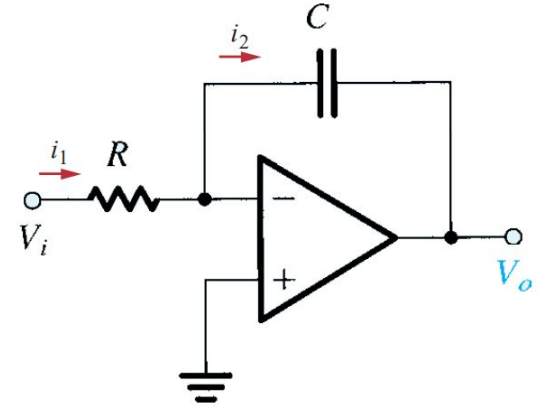
Since there is negative feedback, from virtual short, $v_- = v_+ = 0V$

From Ohm's law for $R \Rightarrow i_1 = \frac{v_i - v_-}{R} = \frac{v_i}{R}$

For the capacitor $C, \Rightarrow i_2 = C \frac{dv_C}{dt} = C \frac{d(v_- - v_o)}{dt} = -C \frac{dv_o}{dt}$

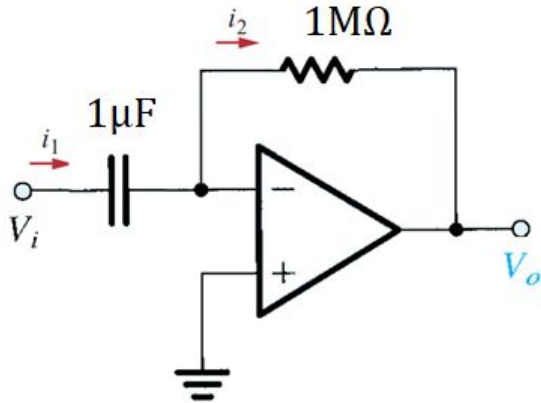
Since ideal op-amp, $i_- = i_+ = 0$, so $i_1 = i_2$

$$\begin{aligned}\Rightarrow \frac{v_i}{R} &= -C \frac{dv_o}{dt} \\ &= -RC \frac{dv_o}{dt} \\ \Rightarrow v_o &= -\frac{1}{RC} \int v_i dt\end{aligned}$$



Example

Observe the following Figure. If $v_i = 5\sin 6t$, Find the value of v_o .



Solution:

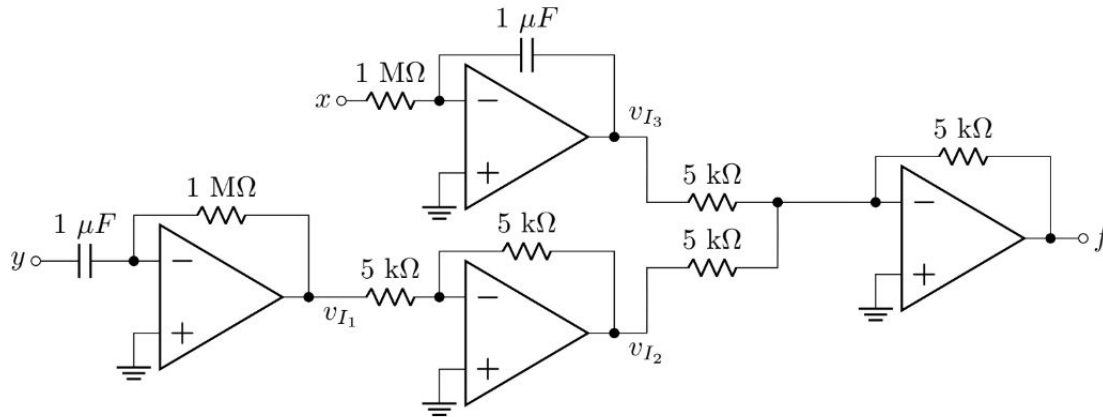
This is a **differentiator**.

$$\text{So, } v_o = -RC \frac{dv_i}{dt} = -1 \times 10^6 \times 1 \times 10^{-6} \times \frac{d(5\sin 6t)}{dt}$$

$$\Rightarrow v_o = -1 \times (5 \times 6 \cos 6t) = -30 \cos 6t \text{ [Ans.]}$$

Example

Analyze the circuit below to **find** an expression of f in terms of inputs x and y .



Solution:

$$v_{f1} = -\frac{dy}{dt}; v_{f2} = -\frac{1}{RC} \int x\ dt; v_{f3} = -v_{f1} = \frac{dy}{dt}; v_o = -(v_{f1} + v_{f2} + v_{f3})$$

Op Amp as Comparator

- A comparator compares two voltages to determine which is larger.

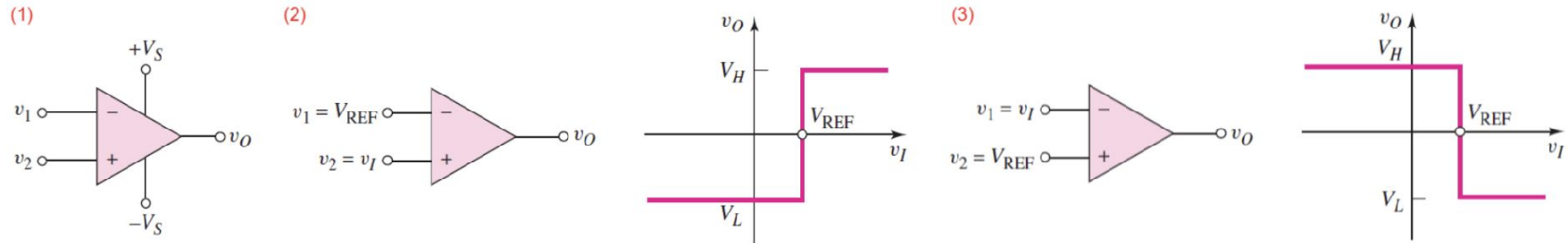


Figure 2: (1) Op-Amp Comparator (2) Noninverting Circuit (3) Inverting Circuit